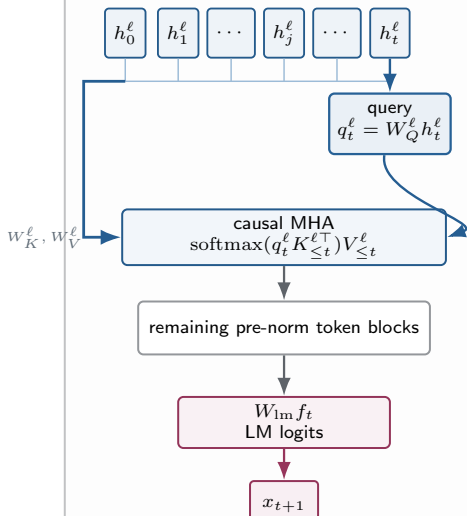


WIKITEXT-2: LOCAL AND GLOBAL CONTEXT PATHS

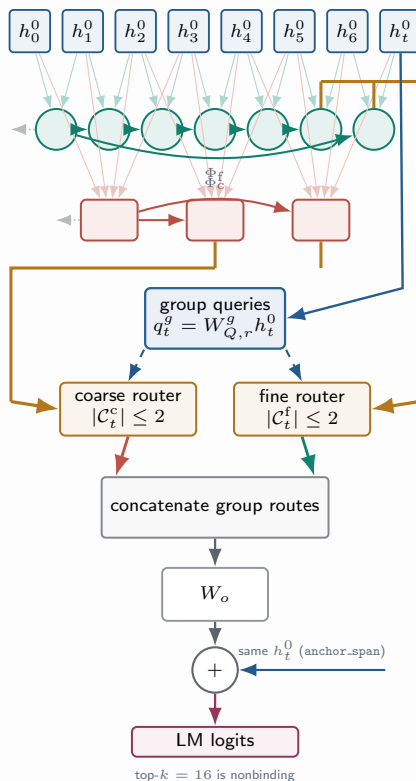
Every non-masked position predicts the next token; each column follows one executed model from states to logits.

Dense token control



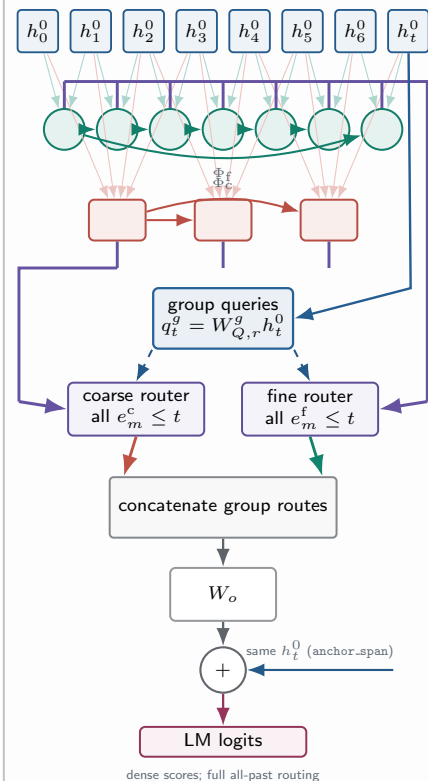
The current state supplies the query; the causal layer states supply keys and values.

Registered local set recipe



top-k = 16 is nonbinding

Global recipe-regression bridge



dense scores; full all-past routing

Shared visual grammar. Every computation runs downward. Upper green circles are fine atoms; lower red rectangles are coarse atoms. A blue query box is a learned projection of the current state, never a merge or decision. Candidate rails lie outside the atom rows: fine above, coarse below. Amber rails admit recent endpoints; violet rails admit every sealed past atom. Rails identify eligible indices: router keys use pre-stack pooled states and routed values use post-stack contextualized atoms.

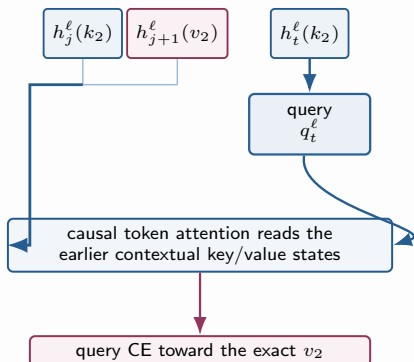
SYNTHETIC MQAR: EXACT BINDING TRANSPORT

A repeated key must recover its earlier paired value; only query-target positions are supervised.



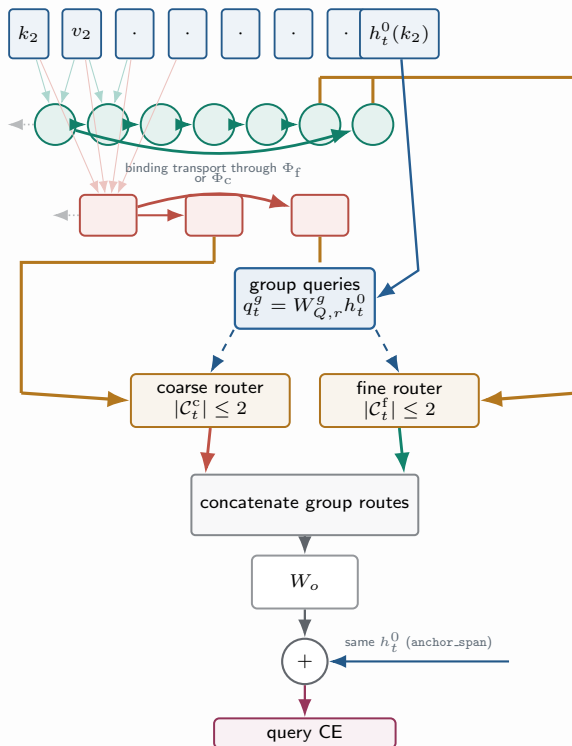
The evidence unit is the earlier binding (k_2, v_2), not a distributed count.

Dense token path



The current repeated-key state forms the query; no recent atom carrier is required.

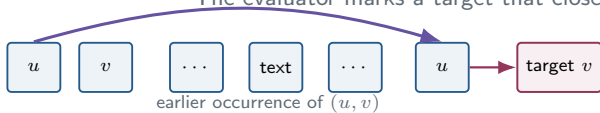
Registered b25 local set path



The remote binding atoms are absent from the amber candidate lanes. The binding can affect the repeated key only if pooling preserves it and the causal atom stack transports it into a recent eligible atom. Top- $k = 16$ is non-binding.

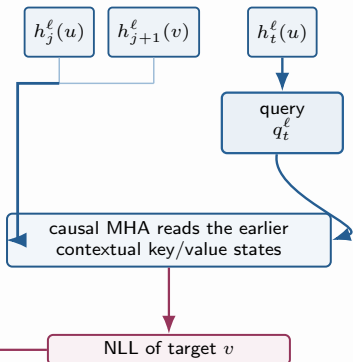
NATURAL REPEATED-BIGRAM AR: FIXED TARGET, DIFFERENT READBACK

The evaluator marks a target that closes an earlier bigram; checkpoints are not retrained.



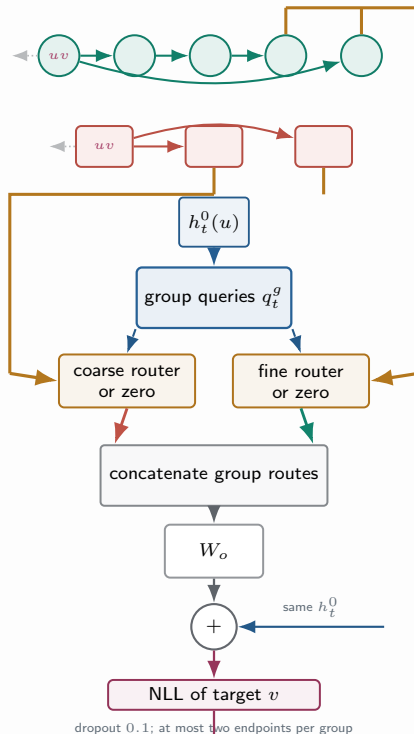
AR-hit is a label derived from the same WikiText-2 sequence; NLL is partitioned after scoring.

Dense token checkpoint

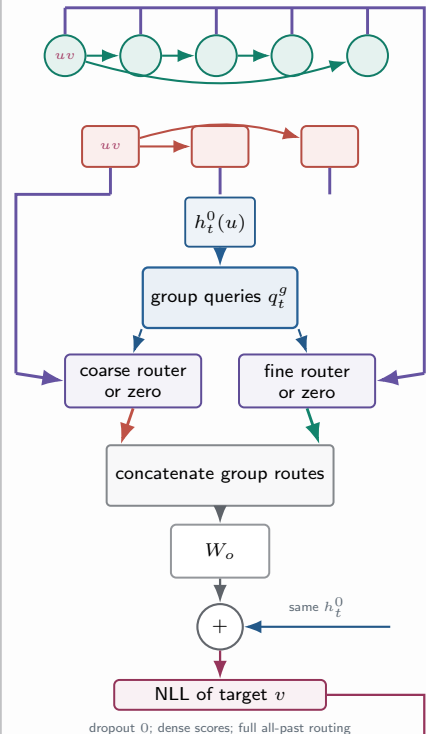


No routed-group ablation.

Registered local set checkpoint



Global-recipe bridge



Identical evaluator: per-sequence AR/non-AR NLL $\rightarrow$ same lag and repetition bins $\rightarrow$ paired 10,000-resample sequence-block bootstrap

Only the contextual path changes. Group removal measures dependence on an allocated fine/coarse route, not a dedicated retrieval circuit.

LCA MARKER COUNTING: DISTRIBUTED EVIDENCE AND READBACK

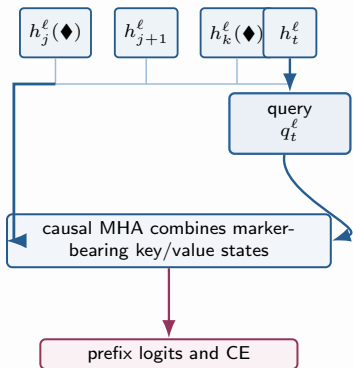
For each supervised prefix, predict whether its marker count exceeds the configured threshold.



Initial calibration supervises only the final t ; the repaired task supervises every valid prefix.

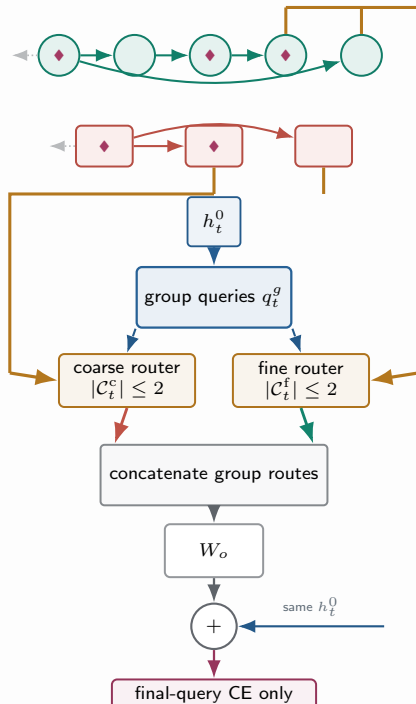
$\blacklozenge$ = marker; N_t counts markers in the prefix

Dense token control



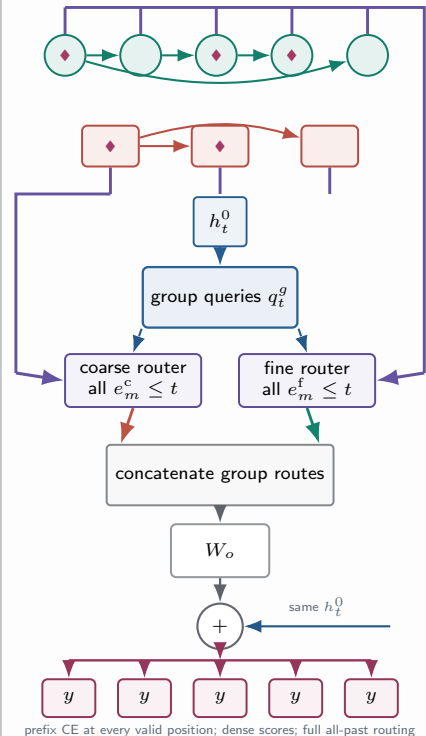
Every query natively reads its full token prefix.

Initial local set calibration



top- $k = 16$ is nonbinding; observed: chance level

Repaired b75 global recipe



prefix CE at every valid position; dense scores; full all-past routing

Three controlled changes. Candidate reachability changes from two recent endpoints to every sealed past atom; routing bandwidth changes from an already nonbinding top- $k = 16$ to full routing; supervision changes from one final target to every valid prefix. Score mode changes memory materialization, not the dot-product/softmax routing equation.